Freeform take: you're Van Nuys Tower

Skyhawk 2SP, a Cessna 172, starts on a right downwind for 16R at KVNY. You work it live over push-to-talk; OBS records the whole thing.

1 · SET UP

1. Headset in, then make it the Mac's mic and speakers:

   ```
   swift scripts/demo/audio_device.swift input HyperX
   swift scripts/demo/audio_device.swift output HyperX
   ```

2. Start X-Plane 12, then size its window to the left half (16:9):

   ```
   swift scripts/demo/xplane_window.swift 0 30 2456 1410
   ```

3. Open the take terminal and drag it to the right half:

   ```
   open -na Ghostty.app --args --window-save-state=never
   --title="sim-pilot take" --font-size=20 --working-
   directory=$HOME/src/an-llm-flies-x-plane-plan
   ```

4. Start OBS, then build the scenes and audio (run it again after restarting X-Plane or the terminal):

   ```
   uv run scripts/demo/obs_setup.py --terminal "sim-
   pilot take"
   ```

   Scenes: Cockpit + Terminal, Terminal + Cockpit, Cockpit, Terminal. Your mic is on track 2, desktop audio on track 3.

2 · RUN

In the take terminal:

```
scripts/demo/take.sh demo/kvny-16r-freeform.yaml
```

It builds, loads the 172 on the downwind, and settles for about 6 s. Then it rolls by itself: OBS starts recording, the terminal opens on the spatial view, and the pilot calls you.

3 · CONTROLS

Tab (hold)	Talk as ATC. Release, check the transcript, then **Enter** to send. Fix "16 right/left" if it's misheard.
Space (hold)	Talk to the pilot off the radio: "how far are we from Burbank?"
Ctrl-V	Spatial view ↔ log. After each plan the code shows for 12 s, then it goes back. Press Ctrl-V yourself and it stays where you put it.
OBS	Switch scenes yourself, from OBS or the phone remote.

4 · STOP

- **Esc** (or Ctrl-C) in the TUI. That ends the take and stops the OBS recording.
- Don't close the terminal window or `kill -9`: the recording would keep going.
- The video goes to `~/Movies`, the take's log and timeline to `output/sim_pilot-<time>.*`.

First take: afterwards, have Claude measure your mic against X-Plane on the recording, and set the gain.

5 · SCRIPT

Wait for each readback before the next call. Say "Two Sierra Papa" and "one six right/left".

A · The fixes

Tower 119.3

After its first call:

> "Skyhawk Two Sierra Papa, Van Nuys Tower, extend downwind, I'll call your base."

Past the 16R threshold:

> "Two Sierra Papa, make a right three-sixty for spacing, then continue downwind."

After the 360:

> "Two Sierra Papa, change to runway one six left, continue downwind."

It should keep waiting for your base call.

20–30 s later:

> "Two Sierra Papa, turn base, runway one six left, cleared to land. Keep it at seventy knots on final, traffic behind you."

Slowing after touchdown:

> "Two Sierra Papa, turn right at Golf, contact Ground one two one point seven."

Ground 121.7

After its call to Ground:

> "Skyhawk Two Sierra Papa, Van Nuys Ground, taxi to Prop Park via Golf, Alpha, cross runway one six right."

Give the crossing in this call: it can't hold short on the way to parking.

B · Landmarks

Tower 119.3

After its first call:

> "Skyhawk Two Sierra Papa, Van Nuys Tower, extend downwind, remain south of the 118, I'll call your base."

About abeam the numbers:

> "Two Sierra Papa, traffic is a jet on a four-mile final. Proceed direct Cal State Northridge and hold over the campus, I'll call you back."

After a lap or two:

> "Two Sierra Papa, traffic no factor. Leave the campus, enter a right base runway one six right, cleared to land."

Then:

> "…turn right at Golf, contact Ground one two one point seven."
> → *Ground:* "…taxi to Prop Park via Golf, Alpha."

No crossing needed from 16R.

Extras to improvise

- "Turn base at Nordhoff" / "extend your downwind to Devonshire, turn base there"
- "Remain east of Reseda Boulevard"
- "Follow the 405 southbound, keep it off your left wing"

Don't call base while it's still beside the runway: it turns right away and the final ends up too short.